



KTU NOTES

The learning companion.

**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPER**

 Website: www.ktunotes.in

Early 1900's

Vacuum Tubes

First generation

1947

Transistors

Second generation

1958

Integrated Circuits
(I.C)

Third generation

1960's

Small scale Integration
(SSI)

Less than 100 components per chip

1966's

Medium scale Integration
(MSI)

100 - 1000 components per chip

1969

Large scale Integration
(LSI)

1975

Very large scale Integration
(VLSI)

More than 1000 components per chip

Ultra large scale Integration
(ULSI)

More than 100,000 components per chip

more than 100,000 components per chip

- arithmetic and logical operations, for solving, making decisions and sharing information.
- 3, Plays a key role in implementation of control systems which are used for variety of applications.
 - 4, plays a major role in domestic and commercial entertainment systems. Eg: Lighting systems, music system etc.
 - 5, Communication systems like radio, television, communication, internet, optical fibre etc.
 - 6) Biomedical appliances and equipment like ECG, X-ray, CT scan.
 - 7, Defence applications Eg: RADAR, sonar, Drones etc.

Active Component

Eg: ① Tube devices like vacuum diode
Vacuum triode etc.

② Some P-N junction diodes
Eg: Tunnel diode, Zener diode.

③ Bipolar junction diode (BJT)

④ FET

⑤ UJT

Passive

Eg: ①

②

③

Active components → Capable of processing an electrical signal. They inject power into a circuit.
Eg: BJT

Passive components → They are not capable of generating or processing energy.

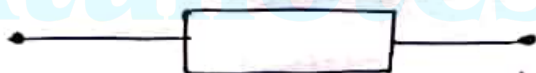
a circuit - Resistors is rated in



Accredited

(American National
Institute)

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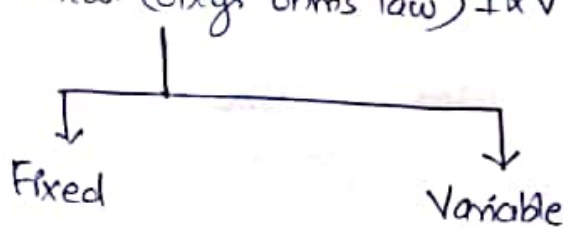
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(International
Commission)

↳ Conductance $G = \frac{1}{\text{Resistance}}$ Unit

↳ ρ = Resistivity (Ωm) → depends on material

↳ Resistance of materials may vary with temperature.



(R varies with linearly)
(1)

(1) Carbon compositions

(1) Potentiometers

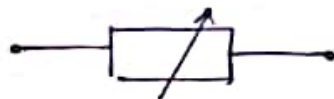
(2) wire wound



(3) Thin film

(2) Rheostats

(4) Carbon film

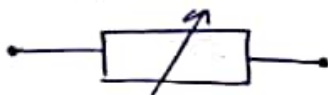


(5) Metal film

(6) Metal oxide

(3) Trimmers

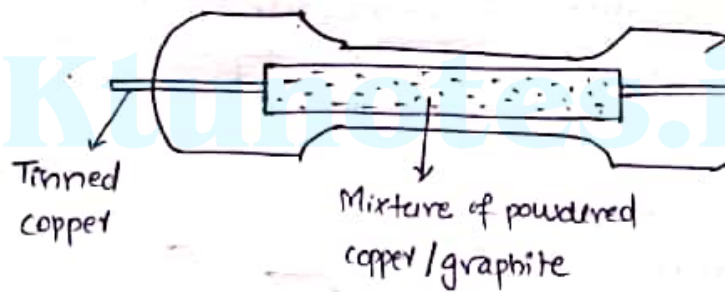
(7) Thick film etc.



* Linear Resistors :

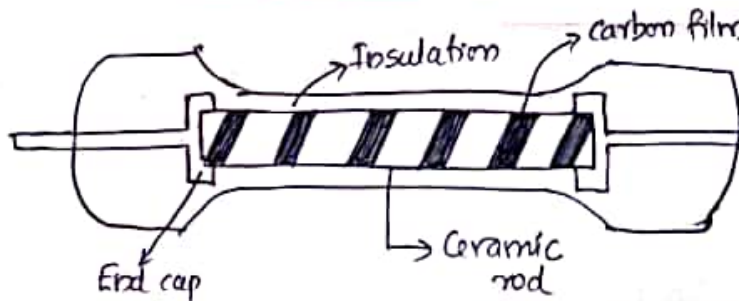
Obeys Ohm's law i.e., At a constant temperature the value of current flowing through is directly proportional to the voltage across
($V = IR$)

↳ Usually ceramic is used as **insulation**
finned copper wires as leads



↳ Small size, rugged, low cost, suitable for frequency applications

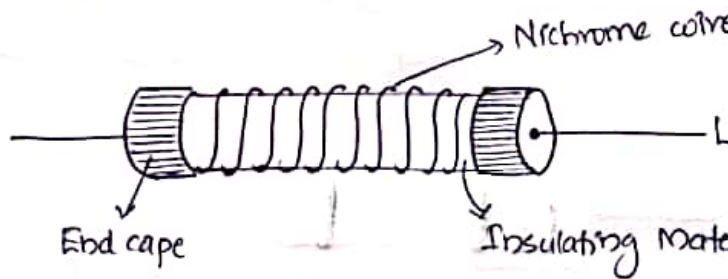
② Carbon film Resistors (10^{-2} to 10^6)



humidity

③ Wire wound Resistors (1- Ω to 200 k- Ω up to 1000 W)

- ↳ Made by winding a wire of alloy around a ceramic or porcelain core
- ↳ can be used for large power applications are larger in size



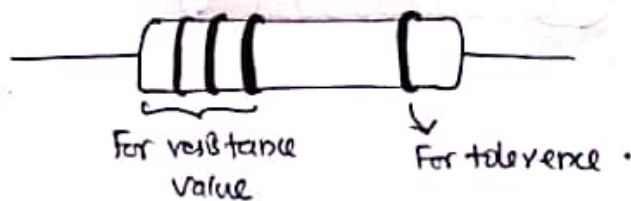
→ Desired resistance obtained by cutting grooves on deposited nichrome film.

*

Colour Coding of Resistors

→ To identify resistance values. Some resistors use colour coding Eg: in film and carbon

→ Four colour band system are used



Yellow	4	4	10^4
Green	5	5	10^5
Blue	6	6	10^6
Violet	7	7	10^7
Grey	8	8	10^8
White	9	9	10^9
Gold	—	—	10^{-1}
Silver	—	—	10^{-2}
No colour	—	—	$\vdots$

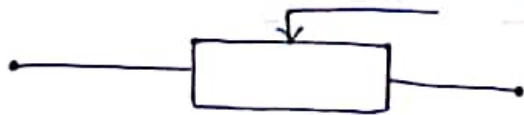
Eg: - If colour code is orange, orange, yellow

I band → Orange → 3	} 33×10^4
II band → Orange → 3	
III band → yellow → 10^4	
IV band → Gold → $\pm 5\%$	

330

① Potentiometers:

- ↳ For low power applications,
- ↳ 3 terminals (one adjustable contact)
- ↳ It is a voltage divided device.
- ↳ Used in volume control on audio eq sticks, Light dimmers etc.



② Rheostat:

- ↳ Two terminal variable resistors
- ↳ Can be used for large power applications (electrical systems eg: welding)



↳ In radio receivers, audio-video components

* Non Linear Resistors

① Thermistor

- ↳ Temperature sensitive resistors
- ↳ Mostly in temperature sensors and heating elements
- ↳ Replacement for fuses, used in electronic for protection.
- ↳ Automotive sensors, eg: Thermocouple chip.



③ LDR - Light Dependent Resistor

- ↳ R varies with, intensity of light
- ↳ Have high $m\Omega$ resistance in dark (no light) when light falls.

↳ Exhibit photo conductivity Eg: On lights
Outdoor
Solar

④ Thyrite : Has high R at low current
at high current. Used in lig

⑤ SMD : Surface Mount Technology

Used in PCB
smaller in size
 η increases
faster

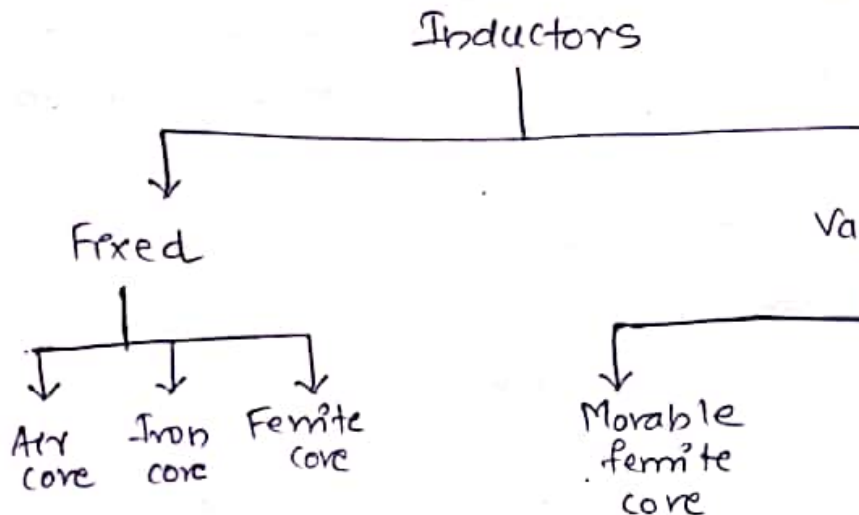
→ It opposes change in current flow
it.

↳ Stores energy temporarily as magnetism

$$\rightarrow V \propto \frac{di}{dt} \quad (\text{or}) \quad V = L \frac{di}{dt} \quad L =$$

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* Types of Inductors



*

Fixed type Inductors

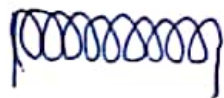
Air core inductor

↳ This type of inductor is made of wire-wound over a former made of board.

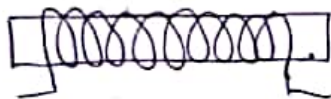
↳ It has a low value of inductance. Suitable for radio-frequency applications.

↳ It can operate at high frequency range to 10 MHz.

↳ Stray field radiation and pickups of electromagnetic interference which become with increase in the diameter of the



- ↳ Laminated core consists of thin iron sheets pressed together and insulated from each other.
- ↳ Iron core inductors are very stable for low frequency applications.
- ↳ They are widely used in audio equipment, industrial power supplies, inverter systems, etc.
- ↳ The core assist in increasing the inductance without increasing the number of turns.

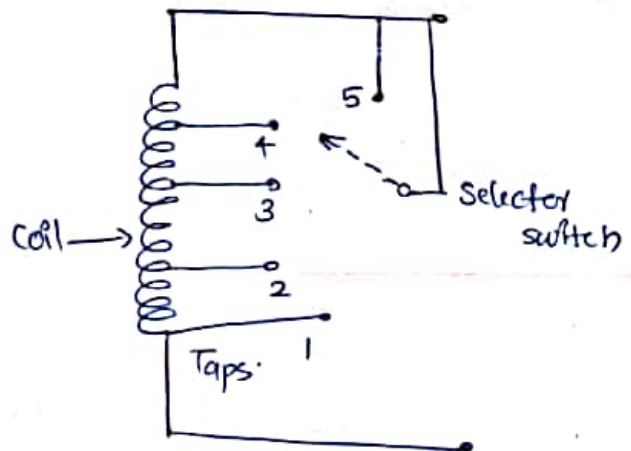


* Ferrite core inductor

- ↳ In this type, a coil of wire is wound on a solid core made of ferromagnetic material called ferrite.
- ↳ In variable type ferrite core inductor, the core is made of a material whose permeability can be varied.

↳ It consists of a coil having large turns and wound on a magnetic core with a desired number of tapings.

↳ Where tap is a conducting wire taken from coil at a desired distance due to different mutual inductance can be obtained over the same inductor.

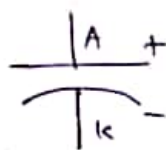
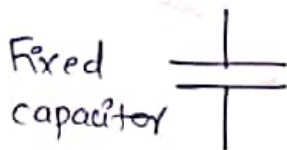


↳ $i = C \frac{dv}{dt}$ where C = capacitance.

↳ They store energy as electrostatic energy.

↳ Usually capacitor of values μF and

Symbol



Polarised capacitor



↳ A capacitor act as open circuit for DC (low frequency) and allows higher signals to pass.

∴ It can be used as a filter.



Capacitance value $C = \frac{\epsilon A}{d}$ Far

Where ϵ = Permittivity of dielectric

A = Area of plate

d = distance between plates.

$\epsilon = \epsilon_0 \epsilon_r$ where ϵ_0 = Perm
Free

$= 8.84$

ϵ_r = Relative

Paper Capacitor
Electrolyte capacitor
Mica capacitors etc.

② Variable capacitors

Ex: 1, Tuning Capacitor

2, Trimmer

3, Padder

* ① Ceramic Capacitor (PF to few μF)

↳ Also called disc capacitor. made
two sides of a ceramic or porcelain
silver.

↳ Can be made very small in size
value E

↳ 3 digit code will be printed on it
value. Eg- ~~103~~ 103 = 10×10^3 pF
= 10×10^3 PF

- ↳ They are usually polarized capacitors, hence not to be used in AC.
- ↳ Usually available in $1\ \mu\text{F}$ to $47\ \mu\text{F}$.

③

Varieg capacitor

- ↳ It is a variable capacitor, capacitor value can be changed mechanically / electronically.
- ↳ They are often used in LC circuits for tuning resonant frequency (radio tuning), capacitor.
- ↳ Air or plastic foil is used as dielectric.

④

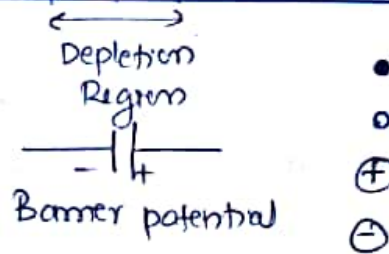
Trimmer capacitor:

- ↳ Small value variable capacitor.
- ↳ Never adjusted after factory setting installation.
- ↳ Air / plastic / ceramic as dielectric.

attracted across the junction to holes in p-side. Thus diffusion takes place across the junction.

↳ Free electrons crossing the junction leave behind negative ions on P side by giving atoms one or more e^- than protons. The electrons also leave positive ions behind them on the n-side. Thus it creates an electric field from N side to P side.

↳ Thus initial diffusion of charge carriers creates a barrier voltage at the junction with negative charge on P side and positive on n side. It is called as space charge potential.



→ N type Semiconductor → Obtained by pentavalent impurity in intrinsic semiconductor

Eg: As, Sb, Pb etc.

There will be free e^- ∴ They are called free electrons impurity.

→ P type semiconductor → Obtained by trivalent impurity in intrinsic semiconductor

Eg: Al, B, Ga, In

It creates extra holes (absence of e^-) and is called acceptor impurity.

is called depletion region. In a P-N junction without external supply a barrier developed across junction and no current through junction.

* VI Characteristics

Forward Biasing

+ve terminal of supply $\rightarrow$ P region

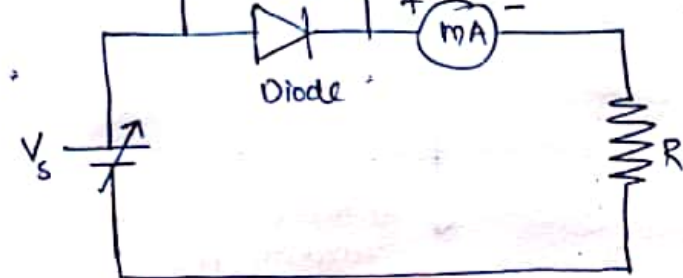
-ve terminal of supply $\rightarrow$ N region

$\hookrightarrow$ When it is forward biased, the holes from +ve terminal of source and it move towards junction. Similarly e^- get repelled from +ve terminal and moves towards junction.

$\hookrightarrow$ Electrons and holes crosses depletion region and recombine themselves. That is width of depletion region decreases.

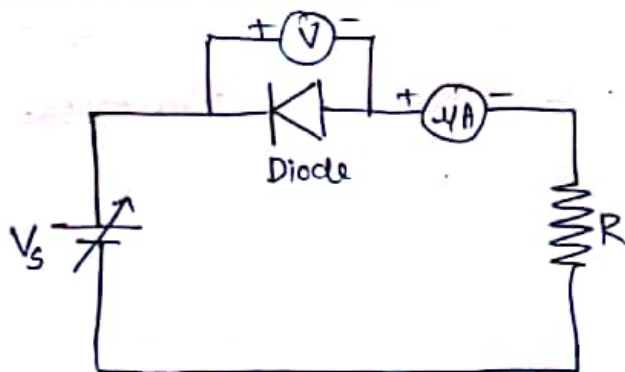
→ Then holes in P region are attracted to the -ve terminal of the voltage source in the 'n' region are attracted towards the +ve terminal.
→ This widens the depletion region and increases the barrier potential. Increase barrier potential prevents the diffusion of majority carrier through the junction.
∴ No current flow due to majority carrier.

→ But the barrier potential helps the minority carriers to cross the junction (holes in n side, e^- in p side) hence small amount of current flows through reverse biased P-N junction.



- ↳ R limits current
- ↳ If we increase voltage then V increase knee value it conducts.

(ii) Reverse biased .



- ↳ If we increase voltage small current (only flows).

Reverse
Voltage

Reverse
current (μA)

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↳ In reverse bias condition, if the applied voltage increases to large values then at some point (sharp increase in current) takes place and this is known as reverse breakdown voltage phenomenon is known as Avalanche

(1) Bipolar Junction transistor (BJT)

(2) Field Effect transistor (FET)

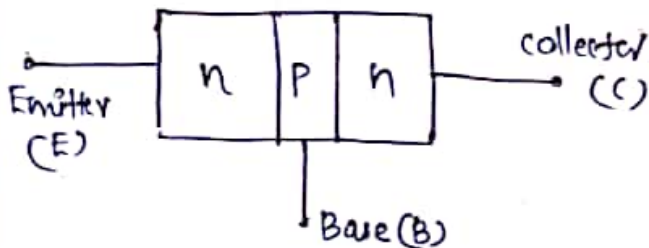
* Construction of BJT

↳ It is a 3 terminal device containing 3 semiconductor layers.

↳ Current conduction is due to both electrons and holes and hence called Bipolar.

↳ There are two different configurations

(1) NPN Configuration



- Emitter $\rightarrow$ Heavily doped compared to supply charge carriers via base. Size more than base but less

- Base $\rightarrow$ Size is extremely small. Carrier from E is directed to collector and only a few recombine in base. Lightly doped

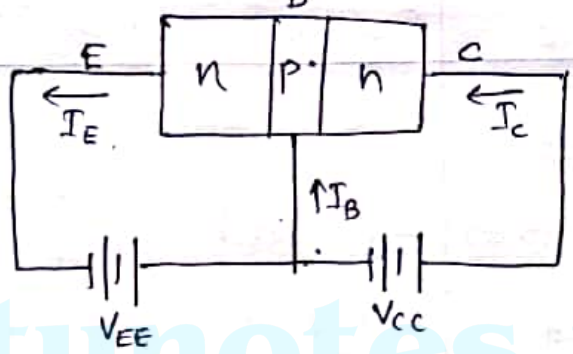
- Collector $\rightarrow$ Moderately doped, Size more than E to collect carriers and to dissipate heat

Active	Forward	Reverse
Saturation	Forward	Forward
Cut-off	Reverse	Reverse
Reverse active	Reverse	Forward

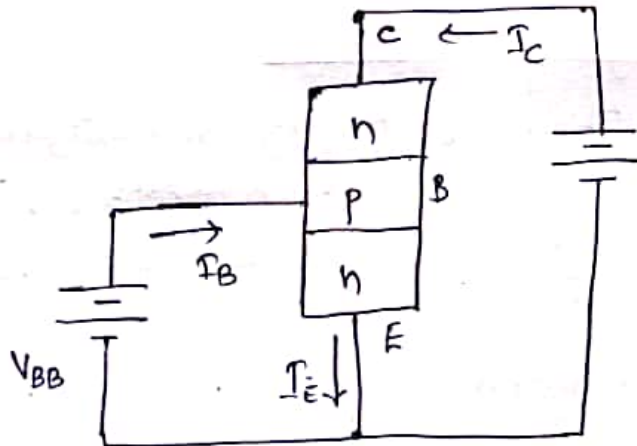
* Configuration of BJT

There are mainly 3 configurations based on its circuit connections, They are

- (i) Common base configuration $\rightarrow$ Base common
- (ii) Common Emitter configuration $\rightarrow$ Emitter common
- (iii) Common collector configurations $\rightarrow$ Collector common



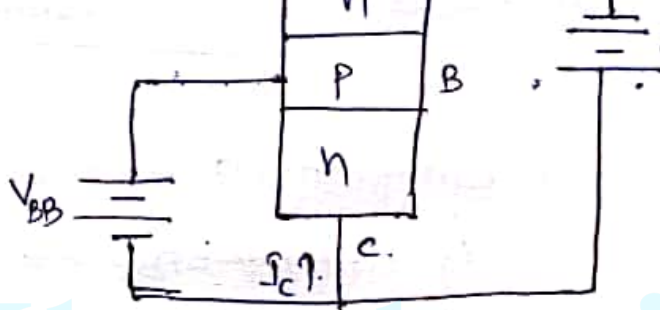
Common emitter configuration $\rightarrow$ In between base and emitter and o/p is c and E



DC current gain

$$\beta = \frac{I_C}{I_B}$$

2021



DC current gain

$$\gamma = \frac{I_E}{I_B}$$

Relation between current gains .

$$\alpha = \frac{\beta}{1+\beta}$$

$$\beta = \frac{\alpha}{1-\alpha}$$

$$\gamma = \beta + 1 = \frac{1}{1-\alpha}$$

voltage
when V_P

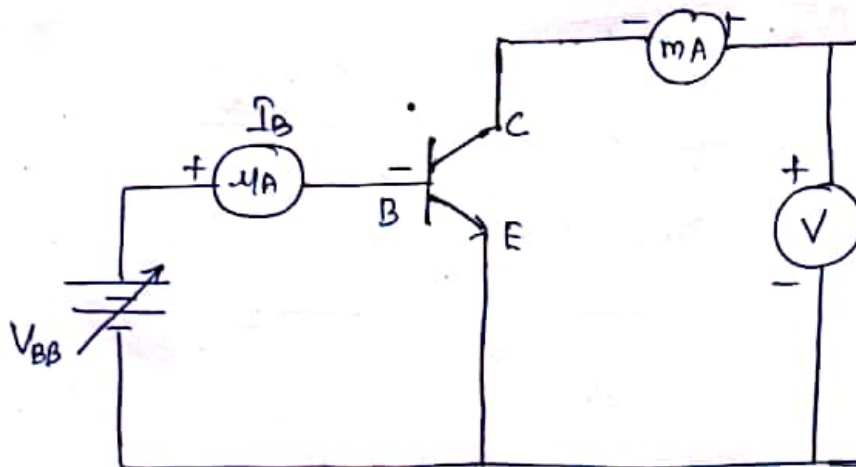
In CE configurations

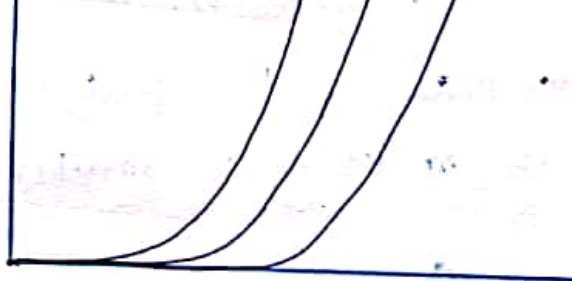
I_P characteristics $\rightarrow V_{BE}$

α constant

O_P characteristics $\rightarrow V_{CE}$ V
for a

$\rightarrow$ The circuit arrangement for obtaining





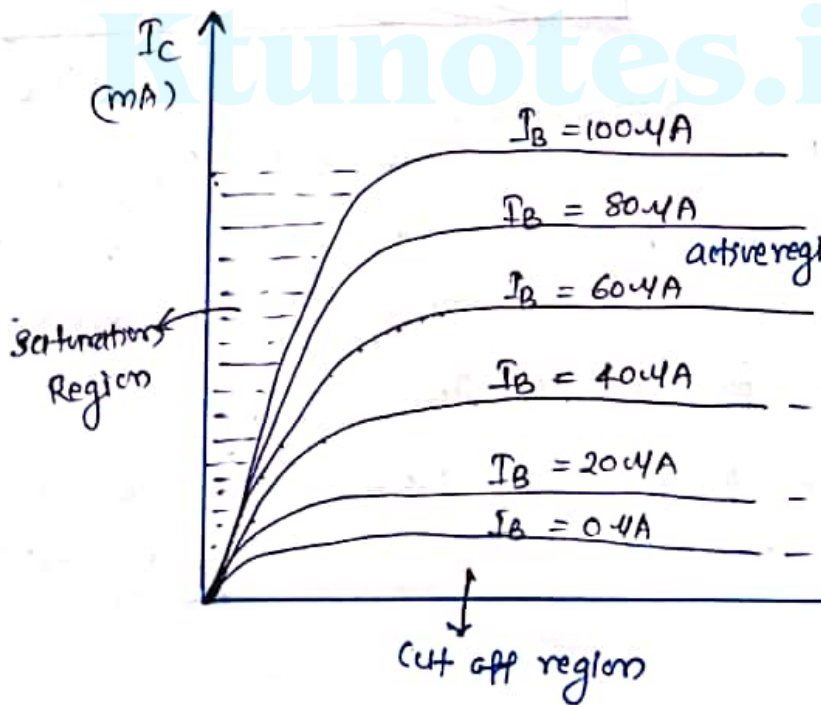
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→ Set V_{CE} as a constant value (say 10V)
 V_{BE} is varied in small steps by changing the base supply and corresponding base current I_B is measured.
Then plot V_{BE} vs I_B

↳ This procedure is repeated for different values of V_{CE} (say 2V, 3V etc) resulting in several curves.

are noted. Then V_{CE} and I_C is plotted

↳ This procedure is repeated for different I_B resulting in similar curves



straight and equally spaced.

↳ The region to the left of V_{CESAT} is the saturation region.

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